HW13

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$2A \xrightarrow{k} \text{Products}$; Dispersed phase mixing
Occurring only on the drops of dispersed — Second order
Liq–Liq dispersion

Cont., CSTR ; $\theta = \frac{\omega N_0 V}{q}$

(i) $\mu_n(\tau) = \int_0^1 x^n \phi(x,\tau)\cdot dx$ ← dimensionless population density

$\mu_0(\tau) = \int_0^1 x^0 \phi(x,\tau)\cdot dx = \int_0^1 \phi(x,\tau)\cdot dx = 1$

(ii) Steady state reactor, CSTR population balance:

$\frac{\partial n(c,t)}{\partial t} + \frac{\partial [\dot{c} n]}{\partial c} = h^+ - h^- + [\text{IN}] - [\text{Out}]$; $h^- = \omega n$

SS

Coalescence – redispersion

$h^+ = 2\omega \iint n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2$

$\dot{c} = -r(c) = -kc^2$

$\frac{\partial [\dot{c} n]}{\partial c} = \omega \iint_{c_1+c_2=c} n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2 + \frac{q}{V}\left(\delta(c-c_0)\right) - \left(\frac{q}{V} n\right)$

$\times \frac{x^r \cdot c_0}{\omega N_0}$ ↓

$\frac{x^r c_0}{N_0}\frac{\partial [-kc^2 \cdot n]}{\partial c} = \frac{x^r c_0}{N_0} \iint_{c_1+c_2=c} n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2 + \frac{q}{V}\delta(x-x_0)\cdot\frac{x^r c_0}{N_0} - \frac{x^r c_0}{\omega N_0}\left(\frac{q}{V} n\right)$

Taking integral $\int_0^1$ & non-dimensionalizing based on class notes:

$n \to \phi$

$x^n \phi \to \mu_n$

[In] term: $\int_0^1 \frac{q}{V} \frac{c_0}{N_0} \frac{\delta(x-x_0)}{\omega}\cdot dx = \frac{q}{\omega V N_0}\int_0^1 \delta(x-x_0)\cdot dx = \frac{1}{\theta}$

[Out] term: $\int_0^1 \frac{x^r c_0}{N_0 \cdot \omega}\cdot\frac{q}{V} n \cdot dx = \frac{q}{\omega V N_0}\int_0^1 x^r \cdot \left(\frac{c_0 n}{N_0}\right)\cdot dx = \frac{1}{\theta}(\mu_n)$ ($\phi(x,\tau)$)

h^+ term: $x^r c_0 \left[1 \quad \ldots \quad 1 \right] \sum \frac{n!}{\quad} \left[\int_0^1 x^{n-r} \quad \int_0^1 x^r \right]$

$\omega V N_0 \int_0$ ...

h^r term : $\frac{x^r c_0}{N_0} \int n_{c_1} n_{c_2} \, dc_1 \, dc_2 = \frac{1}{2^n} \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} \int_0^1 x^{n-r} \, dx \int_0^1 x^r \phi \cdot dx$

(Using Binomial theorem)

h^n term : $\int_0^1 x^n \phi \cdot dx$

$\frac{\partial[-kc^2 n]}{\partial c}$: $\frac{\partial(-kc^2 n)}{\partial(c_0 x)} = \frac{k}{c_0} \frac{\partial c^2}{\partial x} \cdot n - \frac{k c^2}{c_0} \frac{\partial n}{\partial x} = \frac{k c_0 n}{\omega N_0} \int_0^1 x^{n+1} \phi \cdot dx$

$$\frac{k c_0 n}{\omega N_0} \int_0^1 x^{n+1} \phi \cdot dx - 2^{-n} \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} \int_0^1 x^{n-r} \phi \cdot dx \int_0^1 x^r \phi \cdot dx + \int_0^1 x^n \phi \, dx = \underbrace{\frac{q}{\omega N_0 V} \left(1 - \int_0^1 x^n \phi \cdot dx\right)}_{\text{In - out}}$$

$$\frac{k c_0 n}{\omega N_0} \int_0^1 x^{n+1} \phi \cdot dx = \frac{q}{\omega N_0 V} \left(1 - \int_0^1 x^n \phi \cdot dx\right) - \int_0^1 x^n \phi \, dx + 2^{-n} \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} \int_0^1 x^{n-r} \phi \cdot dx \int_0^1 x^r \phi \cdot dx$$

$$K n \mu_{n+1} = \frac{1}{\theta}(1 - \mu_n) - \mu_n + 2^{-n} \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} \mu_{n-r} \mu_r$$

(iii) $n = 1$;

$$K \mu_2 = \frac{1}{\theta}(1 - \mu_1) - \mu_1 + 2^{-1} \left(\frac{1!}{1!\,0!} \mu_1 \mu_0 + \frac{1!}{1!\,0!} \mu_0 \mu_1 \right)$$

$$K \mu_2 = \frac{1}{\theta}(1 - \mu_1) - \cancel{\mu_1} + \frac{1}{\cancel{2}} \left(\cancel{2} \mu_1\right)$$

$$\boxed{\mu_2 = \frac{1}{K\theta}(1 - \mu_1)}$$

(iv) $\mu_{n+1} \to 0$ for large n ; $\theta \gg 1$

$$\mu_n = \int_0^1 x^n \phi(x, z) \cdot dx$$

$n \to \infty$ $x^n \to 0$ since x is a non-dimensional value b/w 0 & 1

$\therefore \mu_n \to 0 \quad \text{as } n \to \infty$

(v) $K = \{0.1, 1, 10\}$

Have to solve for μ_1 ; $\mu_2 = \frac{1}{K\Theta}(1-\mu_1)$

↳ Have to iteratively solve until large n values where μ_n for large n is equal to 0.

$\mu_0 = 1$
$\mu_1 = f(\mu_2)$
$\mu_2 = f(\mu_3)$
⋮
$\mu_{10} = 0$